

as code. This reduces human error and enhances the repeatability of deployment processes.

- **Multi-cloud support:** Terraform supports multiple cloud providers like AWS, Azure, and Google Cloud, making it ideal for managing multi-cloud environments.
- **Version control:** Infrastructure configurations are stored in version control systems like Git, ensuring that infrastructure can be rolled back to any previous state.

Ansible: Automation for configuration and deployment

Ansible is a simple, agentless automation tool that's used for configuration management, application deployment, and task automation. With Ansible, teams can define configurations in human-readable YAML files, which simplifies managing complex infrastructures and environments.

Why Ansible is important for DevOps in 2025

- **Agentless architecture:** Ansible doesn't require agents to be installed on target machines, making it easy to deploy and scale.
- **Configuration management:** It automates server setups, software installations, and updates across multiple machines, ensuring consistency.
- **Playbooks and roles:** Ansible's use of playbooks (sets of instructions) allows for easy automation of tasks like updating configurations or installing applications.

Top open source tools for cybersecurity experts in 2025

As cybersecurity threats evolve and become more sophisticated, security professionals are turning to open source tools to protect their infrastructure, identify vulnerabilities, and monitor networks for potential breaches. The open source community has provided a wide range of tools that help organisations fortify their security posture without relying on expensive proprietary solutions. In 2025, open source tools will continue to be essential in defending against cyber threats, offering powerful, flexible, and cost-effective solutions.

Here are some of the top open source tools that cybersecurity experts should consider using.

OWASP ZAP (Zed Attack Proxy): A comprehensive web application security scanner

The OWASP ZAP (Zed Attack Proxy) is one of the most widely used open source tools for identifying security vulnerabilities in web applications. It acts as a proxy between the browser and the web application, allowing security experts to detect a wide range of vulnerabilities

like SQL injection, cross-site scripting (XSS), and insecure API endpoints.

Why OWASP ZAP is crucial for cybersecurity in 2025

- **Automated vulnerability scanning:** OWASP ZAP can automatically scan web applications for known vulnerabilities, which is vital for continuous security testing in agile development environments.
- **Active and passive scanning:** It offers both active and passive scanning modes, allowing security teams to test applications without disrupting their functionality.
- **Community support:** Being an open source tool, ZAP is continuously updated by the community, ensuring it stays up to date with the latest threats and vulnerabilities.

Metasploit Framework: The penetration testing powerhouse

Metasploit is an open source penetration testing framework that enables security professionals to simulate attacks on systems and identify weaknesses before they can be exploited by malicious actors. It provides a large collection of exploits, payloads, and auxiliary modules that can be used to test the security of applications, networks, and devices.

Why Metasploit is essential for cybersecurity in 2025

- **Comprehensive exploit database:** Metasploit has an extensive database of exploits, which makes it a versatile tool for penetration testers and security researchers.
- **Post-exploitation modules:** After a system is compromised, Metasploit offers modules for maintaining access, escalating privileges, and gathering valuable information about the network.
- **Integration with other tools:** Metasploit can be integrated with other security tools like Nmap, Nessus, and Wireshark, providing a comprehensive security testing platform.

Snort: A powerful network intrusion detection system (NIDS)

Snort is one of the most widely used open source network intrusion detection and prevention systems (NIDS/NIPS). It helps monitor network traffic in real time and identifies malicious activity by inspecting packets for patterns that match known attack signatures.

Why Snort is important for cybersecurity in 2025

- **Real-time monitoring:** Snort can monitor network traffic in real time, enabling security professionals to detect and respond to threats quickly.
- **Flexible and customisable:** Snort can be customised to detect specific threats tailored to the organisation's